
Where the Admissible Radius Binds: a Correction and a Measured Boundary in Online Conformal Prediction

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Abstract

1 A printed claim that the scorecaster breaks online conformal coverage meets its
2 refutation by overlapping authors; this paper joins them and locates the edge
3 of the set that refutation relies on. Conformal PID states long-run coverage for
4 any bounded scorecaster; a corollary keeps it for predictable perturbations of the
5 deployed threshold inside an admissible radius. The claim was acted on; the
6 readout’s placement and the corollary’s derivation are credited by name. At the null
7 scorecaster and the minimal admissible saturator that radius is *tight*. A legal dead
8 band on the *completed* threshold exists that set at its radius; past it miscoverage is
9 1.000000 against $\alpha = 0.1$: coverage fails outright, not merely its rate. The edge
10 belongs to the saturator–scorecaster pair, not the readout; elsewhere the condition
11 is sufficient only. Under the equally legal $\hat{q} \equiv +b/2$ the failing band covers, as
12 measured; at $\hat{q} \equiv -b/2$ both it and partial adjustment at $w = 0.999$ miscover.

13 1 Introduction

14 A refereed paper states that a scorecaster breaks Conformal PID’s coverage guarantee. A later one,
15 by the same first and last authors, proves in the same notation that a bounded perturbation of the
16 deployed threshold does not. The later cites the earlier, so this is ordinary self-correction, and the
17 joining is what this paper adds.

18 An online conformal method [Angelopoulos and Bates, 2023] moves a threshold each round. Forecast
19 stability names, scores and prices that movement [Tunc et al., 2013, Godahewa et al., 2025, Van Belle
20 et al., 2023, Pritularga and Kourntzes, 2024, Genov et al., 2026, Van Belle et al., 2026]. Varying
21 temporal structure at fixed level came earlier, in forecast verification, matching predictive marginals
22 on real producers [Gneiting et al., 2007, Pinson and Girard, 2012, Worsnop et al., 2018]. Coverage
23 and mean length under-describe a deployed interval [Min et al., 2026, Vaze, 2026]. None of it is
24 claimed here.

25 **The record carries a claim and its refutation; this paper joins them.** Conformal PID [An-
26 gelopoulos et al., 2023] adds a *scorecaster* $\hat{q}_{t+1}$ to a saturating error integrator, asking only that it be
27 predictable and lie in $[-b/2, b/2]$: “any function of the past” (Theorem 1). Such a readout is already
28 quantified over. The claim reads “[s]ince using a scorecaster (D-part) in C-PID breaks the theoretical
29 coverage guarantee” [Li and Rodríguez, 2025], stated and acted on in its baselines paragraph, ver-
30 batim on refereed printed page 18443. It is refuted twice, with proof. Li et al. [2025, Corollary 2]
31 add any predictable d_t , $|d_t| \leq \mu_t(b/2 - |\hat{q}_t|)$ with $\mu_t \in [0, 1]$ predictable, to the deployed threshold;
32 since $\hat{q}_t + d_t$ still lies in $[-b/2, b/2]$ it is itself a legal scorecaster, so the result “has exactly the CPID
33 error-integration form” and $|E_T| = o(T)$. The same generic-slot argument carries AcMCP’s long-run
34 coverage [Wang and Hyndman, 2024]. Both identifiers and “scorecaster” were searched across arXiv
35 full text, arXiv abstracts, OpenReview, and Semantic Scholar’s six citers of Li and Rodríguez [2025]:
36 none returns both the claim and the corollary. Hu et al. [2026] still call scorecaster choice “arbitrary”
37 and lacking “principled guidance” while asserting valid coverage: model selection, not validity. **We**
38 **claim neither the placement nor the derivation.**

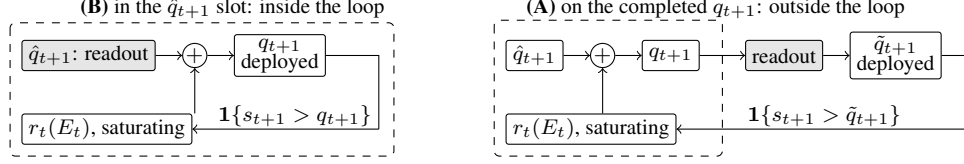


Figure 1: One scalar readout (grey) in its two places in (1); dashed is the loop r_t closes. In (B) the loop closes on the integrator’s own output, and both readouts keep $\{\hat{q}_t\}$ in the convex hull of $\hat{q}_1$ and the raw scorecaster’s range, inside Corollary 2’s admissible set. In (A) it closes on $\tilde{q}_{t+1}$, which it never computed, and nothing keeps the deployed sequence inside that set.

39 **Where Corollary 2 stops, its radius condition is tight.** It is a *retention* result, silent outside that
40 radius, and the question it leaves is whether the radius is sharp. Section 3 exhibits a legal perturbation
41 leaving it: the L_1 dead-band family characterised here on the *completed* threshold, whose edge
42 coincides with the corollary’s radius at the null scorecaster. Past it coverage fails outright, not merely
43 its rate: the published sufficient condition is necessary there. Appendix C sets both claims beside
44 four vocabularies for the same movement.

45 2 Setup

46 **The producer.** A forecaster fit before t induces a score s_t , and a threshold q_t is emitted before
47 y_t is revealed. The deployed set is $C_t = \{y : s_t(x_t, y) \leq q_t\}$ and $\text{err}_t = \mathbf{1}\{s_t(x_t, y_t) > q_t\}$. The
48 target is long-run coverage, $T^{-1} \sum_{t \leq T} \text{err}_t = \alpha + o(1)$, with no model on the data. Conformal PID
49 [Angelopoulos et al., 2023] manipulates the threshold on the score scale:

$$q_{t+1} = \hat{q}_{t+1} + r_t \left(\sum_{i \leq t} (\text{err}_i - \alpha) \right), \quad (1)$$

50 Angelopoulos et al.’s iteration (5). Its r_t saturates: condition (4) requires $|r_t(x)| \geq b$ with the sign of
51 x once $|x| \geq c h(t)$, for h nonnegative, nondecreasing and sublinear. With $E_t = \sum_{i \leq t} (\text{err}_i - \alpha)$,
52 Proposition 2 gives $|E_T| \leq c h(T) + 1$, read on realised scores.

53 **One scalar, two readouts, two places.** Damping a sequentially updated scalar has two standard
54 readouts, both prior work. A quadratic cost gives linear partial adjustment $\tilde{q}_{t+1} = (1 - w)\hat{q}_{t+1}^{\text{raw}} + w\tilde{q}_t$
55 [Gârleanu and Pedersen, 2013], published as post-processing [Godahewa et al., 2025] and applied
56 to a deployed threshold [Binny and Dixit, 2025, Eq. 13]. A proportional cost gives a no-trade band,
57 moving only when the target is more than τ away, then pulling back exactly τ [Constantinides,
58 1986, Davis and Norman, 1990]. Either readout has two positions in (1), drawn in Figure 1; (A),
59 Placement A, is measured here. *The dead band (L_1) is not a stylistic alternative to the partial-*
60 *adjustment (L_2) form;* it is the family whose failure Section 3 characterises. Neither is safe by
61 construction: L_2 forfeits Proposition 2’s rate while covering, L_1 past the radius loses coverage, and
62 at the legal $\hat{q} \equiv -b/2$ both lose it.

63 **The quantity, and the names it already has.** We report the deployed threshold’s *travel*, $\mathcal{T}_H =$
64 $\sum_{t=2}^H |q_t - q_{t-1}|$, after actuator travel in process control. The arithmetic is not new, and the name is
65 a convenience rather than a contribution, used here for one reading only: Appendix C names ten for
66 the same movement across four vocabularies.

67 **Harness.** The harness fixes $\alpha = 0.1$ and $b = 2$. Its saturator $r_t(x) = b \text{clip}(x/(c h(t)), -1, 1)$,
68 with $c = 1$ and $h(t) = \log(t + 2)$, meets (4) with equality. The scorecaster is null and legal, reducing
69 (1) to the pure error integration Proposition 2 addresses, so the readout is the only manipulated
70 variable. Neither choice is neutral: (4) is a lower bound met exactly here, and Theorem 1 permits
71 scorecasters $b/2$ either side of null, predictable but not required to be constant in t . The adversary is
72 specified, not tie-broken. It plays $s_t = \text{clip}(\tilde{q}_t + \varepsilon, \pm b/2)$, $\varepsilon = 10^{-9}$, against the *deployed* $\tilde{q}_t$: the
73 $\varepsilon \downarrow 0$ reading of “sets the score at the threshold”, firing the strict indicator, legal on both sides of
74 Theorem 1’s $[-b/2, b/2]$, so $b/2 + b/2 = b$ is exactly tight. *That specification names one member*
75 *of a class:* any adapted $s_t \in [-b/2, b/2]$ is legal, the constant $s_t \equiv -b/2$ included. Every number
76 in Table 1 (App. A) is measured against the one adversary above; Section 3’s two boundaries are
77 quantified over the class, and they differ.

78 3 The edge of the admissible set, located and proved

79 **The retention result is published, and tested from outside.** The radius restated in Section 1 is
 80 the one Li et al. [2025, Corollary 2] prove, and it *retains* $|E_T| \leq ch(T) + 1$: the radius is theirs. A
 81 downstream partial adjustment of weight w leaves it; Table 1 (App. A) prices that departure, and five
 82 of its eleven arms exceed the bound at $T = 10^6$. The forfeit is a property of the smoothing *strength*,
 83 not Placement A, and it buys something: travel falls from 28,311 to 202. A readout on the completed
 84 q_{t+1} touches neither r_t nor the integrator’s argument, so it cannot put (4) at risk. It delays a correction
 85 already computed, so smoothing the output does not damp E_t but feeds it.

86 **The L_1 dead band leaves the radius, and the edge is a theorem.** The dead band of Section 2 keeps
 87 $|\hat{q}_{t+1} - q_{t+1}| \leq \tau$ at every round, so “inside Corollary 2” reads $\tau \leq \mu_t(b/2 - |\hat{q}_t|)$. Under saturation
 88 the deployed value is pinned τ below the integrator’s ceiling. Outside the radius two levels decide the
 89 answer and condition (4) names only one of them. Write $A_t^+ = \sup_x r_t(x)$ and $A_t^- = \inf_x r_t(x)$ for
 90 the integrator’s extremes, and $\Lambda_t^+ = \inf_{x \geq ch(t)} r_t(x)$, $\Lambda_t^- = \sup_{x \leq -ch(t)} r_t(x)$ for the levels it is
 91 *guaranteed* to reach once the accumulator clears the radius. Condition (4) says exactly $\Lambda_t^+ \geq b$ and
 92 $\Lambda_t^- \leq -b$; that $\Lambda_t^\pm = A_t^\pm$, that the integrator reaches its extremes where (4) puts them, is a further
 93 hypothesis and not one (4) supplies. Set

$$\tau^{*\pm} = \sup_t (|A_t^\pm| \pm \hat{q}_{t+1}) - b/2, \quad \sigma^\pm = \inf_t (|\Lambda_t^\pm| \pm \hat{q}_{t+1}) - b/2. \quad (2)$$

94 **Failure, over the whole admissible class.** If $\tau > \tau^{*+}$ then $\hat{q}_{t+1} + A_t^+ - \tau < b/2$ at every round,
 95 so $\tilde{q}_{t+1} \leq \max(\tilde{q}_t, q_{t+1} - \tau)$ makes $\{\tilde{q} < b/2\}$ absorbing: a release upward lands below $b/2$, and a
 96 hold or a release downward cannot lift the deployed value at all. Against $s_t = \text{clip}(\tilde{q}_t + \varepsilon, \pm b/2)$ the
 97 indicator is then $\mathbf{1}\{\tilde{q}_t < b/2\}$ and every round misses. If $\tau \geq \tau^{*-}$ the constant legal play $s_t \equiv -b/2$
 98 makes $\{\tilde{q} \geq -b/2\}$ absorbing by the mirror inequality, so no round can miss and the realised rate
 99 is 0 rather than α . Either way long-run coverage is lost, with $|E_T|$ linear in T . Nothing here asks $\hat{q}$
 100 to be constant, asks the extremes to be attained anywhere, or asks $A_t^\pm$ to be finite beyond what the
 101 hypothesis already forces. **Retention, keyed to the reach rather than the supremum.** If $\tau \leq \sigma^+$
 102 then $E_{t-1} \geq ch(t-1)$ forces $q_t \geq b/2 + \tau$ and so $\tilde{q}_t \geq b/2$, whence $\text{err}_t = 0$ against *every* legal
 103 adversary; if $\tau < \sigma^-$ the mirror forces $\text{err}_t = 1$ below $-ch(t-1)$. The accumulator cannot then
 104 leave $[-ch(t), ch(t)]$ by more than one round, and $|E_T| \leq ch(T) + 1$: Proposition 2’s own bound,
 105 retained verbatim rather than weakened to $o(T)$. Using condition (4) alone this already certifies
 106 $\tau \leq b/2 + \inf_t \hat{q}_t$ and $\tau < b/2 - \sup_t \hat{q}_t$, which for a constant scorecaster is Corollary 2’s radius at
 107 $\mu_t = 1$.

108 **The two directions meet, with no gap, on the sub-class the five settings inhabit.** If $\hat{q}_t \equiv \hat{q}$, if
 109 $\Lambda_t^\pm = A_t^\pm$, and if neither depends on t , then $\sigma^\pm = \tau^{*\pm}$ and **coverage is retained against every legal**
 110 **adapted adversary if and only if $\tau \leq \tau^{*+}$ and $\tau < \tau^{*-}$.** For a symmetric saturator that is one line:
 111 coverage if and only if $\tau \leq \sup_x r_t(x) - b/2 - |\hat{q}|$, with the endpoint included only when $\hat{q} < 0$.
 112 Every setting run here is in that sub-class, so the sharp form applies to all five. **What was printed**
 113 **before is the failure half.** τ^{*+} is the rule Figures 2 and 3 (App. A) draw, and all 19 dead-band runs
 114 agree with it read against the adversary the harness plays. It is not a retention condition, it omits τ^{*-}
 115 entirely, and Table 2 (App. A) separates the two with legal objects only. The supremum can sit where
 116 the dynamics never go: since $|E_t| \leq (1 - \alpha)t$, a saturator equal to the harness’s inside $|x| \leq t^2$ and to
 117 $\pm 10b$ outside it satisfies (4), reads $\tau^{*+} = 19$ and fails at $\tau = 1.001$; and a legal time-varying $\hat{q}$ splits
 118 $\sup_t$ from $\inf_t$, $\hat{q}_t = b/2$ on t a power of two and 0 elsewhere reading $\tau^{*+} = 2$ against $\sigma^+ = 1$.

119 **The sharp form is one number, and its hypotheses are what buy it.** Specialise to a symmetric
 120 saturator that meets (4) with equality and attains its extremes there ($\Lambda_t^\pm = A_t^\pm = \pm b$; the harness’s
 121 $r_t = b \text{ clip}(x/(ch(t)), \pm 1)$ is one, Section 2), and to a constant $\hat{q}$. Then

$$\min(\tau^{*+}, \tau^{*-}) = b/2 - |\hat{q}|, \quad (3)$$

122 with the endpoint admissible only when $\hat{q} < 0$: exactly Corollary 2’s radius at $\mu_t = 1$, now as an
 123 if-and-only-if. **The necessity direction above assumes none of it:** $\tau > \tau^{*+}$, or $\tau \geq \tau^{*-}$, forfeits
 124 coverage against every admissible r_t and every legal predictable $\{\hat{q}_t\}$, extremes attained or not, and
 125 that unrestricted statement is what this section claims. The two hypotheses are load-bearing rather
 126 than decorative: the unattained supremum and the varying- $\hat{q}$ split two paragraphs above are exactly

what they rule out. Equation (3) is where the reach meets the supremum, not evidence that they
 always do.

The mirror boundary is already measured, on this paper’s own grid. The τ -cliff run’s exact-
 threshold column, read against the mirror adversary $s_t \equiv -b/2$ (Figure 2, App. A), locates $\tau^{*-} = 1$
 at the null scorecaster, closed on the failing side, exactly where (2) puts it. Both 1.000000 and
 0.000000 are failures of the same order, $|E_T| = (1 - \alpha)T$ and αT , **so the failure criterion is $|E_T|$,**
not the miscoverage rate; and once $\hat{q}$ varies the rate stops being a tell at all.

Both degenerate edges, and what they need. An unbounded supremum is not on its own enough
 for $\tau^{*+} = \infty$ to mean what it looks like: the same unreachable-set construction with $r_t(x) = x$
 past $|x| > t^2$ has no finite supremum and fails at $\tau = 1.5$. What the tangent family has, and that
 construction does not, is a genuine reach: $\Lambda_t(X) = \tan(\arctan(b)X/(ch(t)))$, so the accumulator
 level at which the deployed threshold is forced to $b/2$ is $\kappa(\tau)ch(t)$ with $\kappa(\tau) = \arctan(b/2 +$
 $\tau - \hat{q})/\arctan(b)$, finite at every finite τ . Coverage is retained at every width, as measured; but
 $\kappa(\tau) > 1$ once $\tau > b/2 + \hat{q}$, so Proposition 2’s own *constant* is forfeited past $\tau = b/2$ while coverage
 is not. At $\tau = 1.5$, $T = 10^6$ the excursion is 15.10 against Proposition 2’s 14.8155 and against
 $\kappa(1.5)ch(T) + 1 = 15.8530$. At the other edge, $\sup_t \hat{q}_t = -b/2$ forces $\hat{q} \equiv -b/2$, since every $\hat{q}_t$
 is at least $-b/2$ already. There $q_{t+1} - \tau \leq -b/2 + b - \tau < b/2$ for every $\tau > 0$ and every t , and
 $\hat{q}_1 = 0 < b/2$, so the failing set is absorbing from round 1 with no transient: the admissible window
 is the single point $\tau = 0$, and $\tau = 0.5$ returns 1.000000, as does partial adjustment at $w = 0.999$.

What that establishes is that the published condition is tight. At $\mu_t = 1$, $\hat{q} \equiv 0$, $\tau = 0.9$
 covers at 0.100012 while $\tau = 1.5$ returns 1.000000 with $\max_t |E_t| = 900,000$, 60,747 times
 Proposition 2’s bound at $T = 10^6$ (Table 1, App. A); (2) makes that an if-and-only-if against the
 whole legal adversary class, not a pattern across five settings. **The claim is that necessity: a tightness**
result about someone else’s sufficient condition, smaller and more precise than a forfeit of the rate
 by post-processing. Elsewhere it stays sufficient only, and now provably so: the tangent integrator
 above forfeits no coverage at any width Corollary 2 permits. At $\hat{q} \equiv +b/2$ the looseness goes the
 other way and disappears: Corollary 2 permits radius 0 and (2) no width at all against the mirror
 adversary, the unsmoothed control included, though $\tau = 1.5$ covers there against the specified one.
 That asymmetry makes “tight” a located claim.

The boundary read as a bet. The boundary has a betting reading (Appendix B): under a
 Bernoulli(α) null on err_t , imposed for that reading alone and on nothing else here, an anytime-
 valid test convicts the $\tau = 1.5$ arm past the edge at the 5% level and cannot reject inside it at any
 horizon. It adds a diagnostic and a reading, not a result, and no result above assumes that null or is
 certified by that test.

4 Limitations

The retention direction is proved with no gap only where the two boundaries meet: a constant
 scorecaster and a saturator attaining its extremes at the condition-(4) radius, which is every setting run
 here; a genuinely time-varying $\hat{q}$ leaves the band $\sigma^+ < \tau \leq \tau^{*+}$ open in neither direction (Section 3).
 And the excursion is specific to the saturator’s *level*, which condition (4) bounds only from below.
The inherited guarantee is weak where it applies. Under the tangent integrator $h(t) = t/\log t$,
 the certified gap $(ch(T) + 1)/T$ decays as $O(1/\log T)$ rather than $O(1/T)$. What is forfeited was
 not a tight certificate. **The dead-band widths are absolute, and the failure is total only against**
the adversary. τ is in score units with $b = 2$, so the failing $\tau = 1.5$ is $0.75b$. Under i.i.d. scores it
 returns 0.249376 at $T = 10^6$, an adversary-free confirmation of the mechanism rather than a retreat.
 The threshold pins at $b - \tau = 0.5$, and uniform scores on $[-b/2, b/2]$ put $P(s > 0.5) = 0.25$ exactly.
The harness was rebuilt against numbers already read, which carries less weight than a rerun done
 without sight of them. **Placement B is conceded, not tested. The i.i.d. check above used one seed**
(20260820, $T = 10^6$); the primary, adversarial harness uses none, and only its null-scorecaster
edge is bracketed; the others are one-sided. **What remains open is the band between the two**
boundaries. For a genuinely time-varying $\hat{q}$, coverage in $\sigma^+ < \tau \leq \tau^{*+}$ is neither proved nor
 falsified; Table 2’s power-of-two rows sit there, and closing that band is future work, not a sweep this
 paper has already run.

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282 Code and data

283 The simulator, its frozen configuration, the results/ files behind every figure and table, the gen-
284 erators that read them, and this project’s audit trail across sessions are at [https://\[REPOSITORY](https://[REPOSITORY LINK - INSERTED BY AUTHOR PRIOR TO SUBMISSION])
285 [LINK - INSERTED BY AUTHOR PRIOR TO SUBMISSION\]](https://[REPOSITORY LINK - INSERTED BY AUTHOR PRIOR TO SUBMISSION]).

286 A Supporting measurements

Table 1: Placement A: a one-scalar readout on the *completed* threshold, under one deterministic adversary, on one pass to $T = 10^6$; all eleven arms run are listed. Proposition 2’s bound is 13.2061 at $T = 2 \times 10^5$ and 14.8155 at $T = 10^6$; “ratio” is $\max_t |E_t|$ over it, and “Sat.” the fraction of rounds on which (4) delivers its full $\pm b$. Every number is measured against the adversary Section 2 specifies.

Readout on the completed threshold	miscov. 2×10^5	miscov. 10^6	$\max_t E_t $ 10^6	ratio	Sat. 10^6	travel 10^6
none (control)	0.100035	0.100007	7.80	0.53	0.0000	28,311
partial adj. $w = 0.5$	0.100030	0.100007	7.70	0.52	0.0000	18,113
partial adj. $w = 0.9$	0.100030	0.100007	7.50	0.51	0.0000	4,081
partial adj. $w = 0.99$	0.100040	0.100006	63.00	4.25	0.0007	1,023
partial adj. $w = 0.999$	0.100035	0.100004	623.70	42.10	0.0097	202
growing time constant $p = 0.5$	0.100030	0.100006	12.60	0.85	0.0000	330
growing time constant $p = 0.9$	0.100030	0.100024	53.00	3.58	0.3181	16
running mean	0.099940	0.100010	49.50	3.34	0.3647	9
dead band $\tau = 0.5$	0.100035	0.100005	10.60	0.72	0.0000	2,045
dead band $\tau = 0.9$	0.100060	0.100012	13.20	0.89	0.0038	1,051
dead band $\tau = 1.5$	1.000000	1.000000	900,000	60,747	1.0000	0.5

Table 2: Legal $(r_t, \hat{q})$ configurations on which the printed τ^* over-certifies, each under two legal adversaries. “Proved window” is where (2) certifies retention against every legal adversary; between it and the failure boundaries $\tau^{*\pm}$ neither direction is claimed, and the rows built on a supremum the dynamics cannot reach or never attain sit there. Here $\emptyset$ marks a configuration for which no width is certified, $\tau = 0$ included. $T = 10^5$ except the two power-of-two rows, at $T = 10^6$. The last row carries no dead band at all.

$(r_t, \hat{q})$ configuration	τ	printed τ^*	proved window	$s_t = \hat{q}_t + \varepsilon$		$s_t \equiv -b/2$	
				miscov.	$\max_t E_t $	miscov.	$\max_t E_t $
$\hat{q} \equiv +0.4$, level b	0.599	1.4	$\tau < 0.6$	0.100010	7.20	0.099890	11.60
$\hat{q} \equiv +0.4$, level b	0.8	1.4	$\tau < 0.6$	0.100050	8.70	0.000000	10,000
$\sup_x r_t = 10b$, off the reach	1.001	19	$\tau < 1$	1.000000	90,000	0.000000	10,000
$\sup_x r_t$ unbounded, off the reach	1.5	∞	$\tau < 1$	1.000000	90,000	0.000000	10,000
$A^+ = 4b$, $A^- = -b$	1.5	7	$\tau < 1$	0.100020	4.30	0.000000	10,000
level $4b$, $\hat{q} \equiv 0$	6.999	7	$\tau < 7$	0.100070	12.20	0.100050	11.60
level $4b$, $\hat{q} \equiv 0$	7.0	7	$\tau < 7$	0.100090	12.20	0.000000	10,000
$\sup_x r_t = 2b$, unattained	2.99	3	$\tau < 1$	0.100320	211.10	0.098010	210.50
$\sup_x r_t = 2b$, unattained	3.0	3	$\tau < 1$	1.000000	90,000	0.000000	10,000
$\sup_x r_t = 2b$, attained	3.0	3	$\tau < 3$	0.100000	12.20	0.000000	10,000
$\hat{q}_t = b/2$ on powers of 2	1.0	2	$\emptyset$	0.100009	14.30	0.000000	100,000
$\hat{q}_t = b/2$ on powers of 2	1.9	2	$\emptyset$	0.110960	84,745.70	0.000000	100,000
$\hat{q} \equiv +b/2$, no band	0	2	$\emptyset$	0.100000	0.90	0.000000	10,000

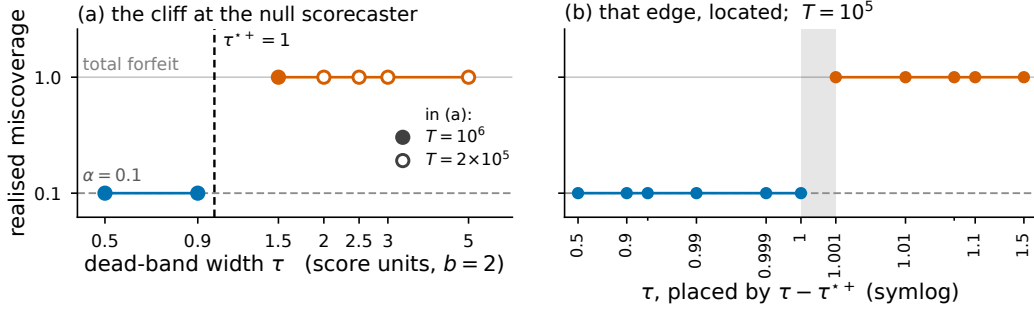


Figure 2: Realised miscoverage against dead-band width τ at the null scorecaster, $\hat{q} \equiv 0$ with a saturator of level b , under the *specified* adversary, where the upward boundary τ^{*+} of (2) is 1. **(a)** The full sweep: filled markers run to $T = 10^6$ and the four widest bands, drawn open, to $T = 2 \times 10^5$; blue covers, orange forfeits, and the dashed rule is τ^{*+} . **(b)** The same setting at $T = 10^5$ over eleven measured widths, placed by $\tau - \tau^{*+}$ on a symlog axis whose linear zone is $|\tau - \tau^{*+}| \leq 10^{-3}$, which magnifies the neighbourhood of τ^{*+} : visual spacing here is not proportional to true distance in τ . Unlabelled minor ticks mark $\tau = 0.95$ and 1.05 . The grey band spans the widest covering width to the narrowest failing one; Figure 3 gives the other four $(r_t, \hat{q})$ settings. The same eleven widths under the *mirror* adversary, which locates τ^{*-} , are this run’s exact-threshold column read under the strict indicator.

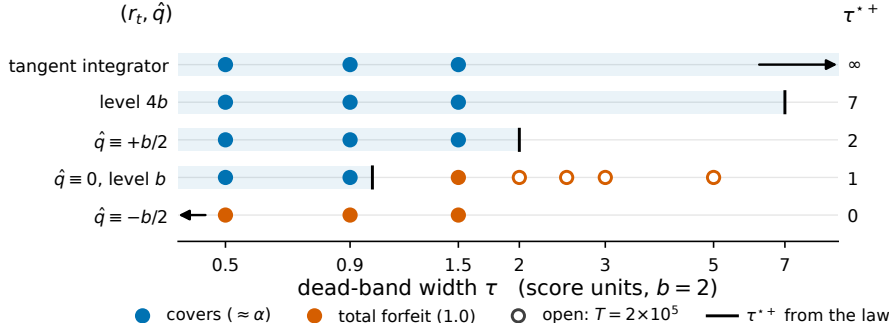


Figure 3: All 19 dead-band runs, at the five admissible $(r_t, \hat{q})$ settings: rows are settings ordered by τ^* , and the shared horizontal axis is τ . Blue markers cover at $\approx \alpha$ and orange markers return 1.000000; filled markers run to $T = 10^6$ and open markers to $T = 2 \times 10^5$, all under the *specified* adversary. The rule on each row is that setting’s upward boundary $\tau^{*+} = \sup_x r_t(x) + \sup_t \hat{q}_t - b/2$, printed at the right, and the tinted band is $\tau \leq \tau^{*+}$. Two rules fall off the axis and are drawn as arrows: $\tau^{*+} = 0$ at $\hat{q} \equiv -b/2$, and τ^{*+} unbounded under Angelopoulos et al.’s tangent integrator. The mirror boundary τ^{*-} of (2) is not drawn here and differs from τ^{*+} at the two non-null scorecasters.

287 B The betting reading

288 The boundary is stated in $|E_T|$, which is the net gain of a Skeptic who pays α each round for err_t ;
 289 that is what gives it a betting reading. The reading Section 3 points at is Ville’s inequality applied to
 290 the standard martingale $M_t = \prod_{i \leq t} (1 + \lambda(\text{err}_i - \alpha))$, whose logarithm depends on the path only
 291 through t and E_t [Shafer and Vovk, 2019, Ramdas et al., 2023, Vovk and Wang, 2021], under the
 292 imposed null that $\text{err}_t \mid \mathcal{F}_{t-1}$ is Bernoulli(α), with $\mathcal{F}_{t-1}$ the history through round $t - 1$. Here λ
 293 is fixed in $[-1/(1 - \alpha), 1/\alpha]$, and its sign is chosen after the fact, not pre-registered. So read, an
 294 anytime-valid test of the deployed intervals’ calibration convicts the $\tau = 1.5$ arm on its fifth interval
 295 at the 5% level, and inside the edge the same test cannot reject at that level at any horizon. M_t is an
 296 e-value for that imposed null alone; the accumulator itself is signed and is neither an e-value nor an
 297 e-process, and no result in Section 3 assumes that null or is certified by that test.

C Where this sits

One quantity, four names. Four literatures name the movement of a sequentially updated forecast in four vocabularies: *abrupt threshold changes*; *forecast (in)stability* (MASC, MAC/SDC, MQC/SMQC, congruence, nervousness); *jumpiness* [Zsótér et al., 2009], *convergence index* [Ehret, 2010], *(in)consistency* [Pappenberger et al., 2011]; and *switching cost*. The same arithmetic is the Multi-Quantile Change over a predictive distribution [Zanotti, 2026, v3, §3.4.1], IACER in applied control [Sarhadi, 2025], and a path length in dynamic-regret online learning. Online conformal prediction gives distribution-free long-run coverage for any bounded scorecaster [Angelopoulos et al., 2023, Thm 1, Prop. 2], generalizing the running-error feedback that adaptive conformal inference [Gibbs and Candès, 2021] and its distribution-shift variant DtACI [Gibbs and Candès, 2024] already apply directly to the threshold. Forecast stability treats the movement as a functional, read out as post-processing [Godahewa et al., 2025, Eq. 3], against a priced decision to switch [Genov et al., 2026, §4.2], [Van Belle et al., 2026]. Verification work uses a matched-level design against real producers from 2012 [Pinson and Girard, 2012, Worsnop et al., 2018]. Switching-cost online learning gives worst-case theory for the penalised problem [Andrew et al., 2013, Thm 2].

The object that settles a deployed interval’s validity (the admissible set around the deployed threshold) sits in the first of them. Those four vocabularies are where this paper’s claim and its concessions have to be placed. **What is conceded, by name.** The placement is prior work: Angelopoulos et al. [2023, Thm. 1, Prop. 2] state coverage, with a rate, for any admissible integrator and any scorecaster, and run a smoothing-family model in that slot themselves; this paper proves that admissible set tight. Dupuy et al. [2026, Appendix A] give the generic argument, and Duerst et al. [2024] adopt that scorecaster. The derivation is prior work too, and this concession matters most. Li et al. [2025, Corollary 2] prove the retention statement in Conformal PID’s own notation, with Wang and Hyndman [2024] an independent second instance. Section 2 concedes the smoother object, the metric arithmetic and both readout maps. Zheng et al. [2025], Wu et al. [2025], Chen et al. [2023], Lade et al. [2026] each place a smoother inside their own validity argument. One vocabulary out, this is integrator anti-windup with a feedforward path: free-until-the-bound-binds is its defining specification, not a finding [Galeani et al., 2009]. Nearest in the other direction, Gao et al. [2025] run the same recursion but gate the *model* update on the conformal score. Their one non-standard assumption is that $\sum_{t \leq T} q_t \leq O(\alpha h(T))$ for a sublinear h , supported with a plot rather than derived. Verification’s matched-level design tests real producers from 2012, with a control stronger than the pair: the whole marginal [Pinson and Girard, 2012, Worsnop et al., 2018]. Switching-cost online learning adds a Thm. 7 trade-off in which sublinear regret is bought by growing θ with T , which grows the ratio [Andrew et al., 2013, Thm. 7].